

HVAC Airflow & Static Pressure Measurement

Pressure quick reference. Use current equipment and instrument procedures; measurement boundaries vary by manufacturer and configuration.

PRESSURE TERMS

- Static pressure**
Pressure in the air system apart from the kinetic contribution represented by velocity pressure.
- Velocity pressure**
Pressure associated with air motion; pitot velocity pressure is total minus static pressure.
- Total pressure**
Static pressure plus velocity pressure at the same point.
- Pressure drop**
Pressure difference across a component or section at a stated operating condition.
- TESP**
Total external static pressure across the equipment's defined external boundary.
- Available static**
Design pressure budget left for distribution losses after applicable external component losses.

CONCEPTUAL TESP RELATIONSHIP



Conceptual only. Approved probe locations and the rated boundary differ among furnaces, fan coils, air handlers, packaged equipment, filters, coils, and accessories.

ILLUSTRATIVE TESP ARITHMETIC

RETURN
-0.20 in. w.c.

+

SUPPLY
+0.35 in. w.c.

=

MAGNITUDE
0.55 in. w.c.

Add magnitudes for this conventional negative-return/positive-supply example. The values demonstrate arithmetic only - they are not normal, recommended, acceptable, code-compliant, or a design target.

PRESSURE CONVERSION

1 in. w.c. = 249.0889 Pa

in. w.c. and in. w.g. are equivalent water-column/water-gauge usage in this reference and the existing AnyHVAC tools.

MANOMETER AND PROBE MINI REFERENCE

INSTRUMENT	USEFUL FOR	WATCH
Digital manometer	Static, differential, component drop, TESP	Range, zero, tubing, stability, calibration
Static probe	Static pressure at an approved test port	Orientation, turbulence, location, leaks
Pitot-static tube	Total/static pressure and duct traverses	Alignment, density, low pressure, point coverage

SAFE AND RESPONSIBLE USE

- Follow current manufacturer procedures. De-energize equipment before creating test ports. Confirm locations are clear of coils, wiring, tubing, controls, and rotating components. Restore ports after use. Document operating mode, fan setting, filter, coil, damper, and accessory condition.
- TESP is not airflow.** Use the correct manufacturer fan data and known operating conditions.
- Measured TESP is not design available static pressure.** Do not enter it directly as Fan External Static Pressure without a valid design workflow.

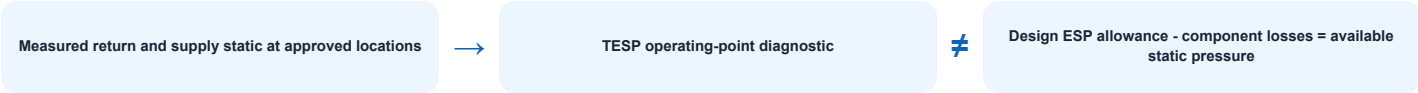
TESP MEASUREMENT WORKFLOW

1. Identify exact equipment and obtain current manufacturer instructions.
2. Define which components are inside or outside the rated boundary.
3. Choose only procedure-approved return and supply test locations.
4. Zero the manometer and connect the static probe correctly.
5. Run the required stable mode and airflow setting.
6. Record both readings and material operating conditions.
7. Compare the magnitude with the correct manufacturer data.

RECORD WITH EVERY READING

- Equipment model
 - Fan or airflow setting
 - Filter condition
 - Damper positions
 - Instrument and range
 - Observed instability
- Operating mode
 - Probe locations
 - Coil condition
 - Installed accessories
 - Calibration status
 - Date and technician

KEEP THE FIELD READING AND DESIGN PRESSURE BUDGET SEPARATE



Airflow Measurement & Calculator Connections

Airflow estimates are only as reliable as the representative velocity, actual free area, instrument, location, and method used.

CORE RELATIONSHIPS

Airflow
CFM = FPM x ft2

Velocity
FPM = CFM / ft2

Rectangular area
A = W x H

Round area
A = pi x D2 / 4

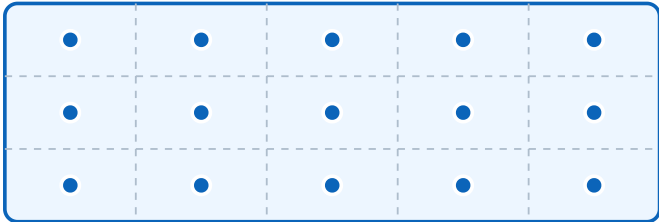
Use actual internal dimensions. Internal liner reduces free area. Convert inches to feet before calculating ft2.

ILLUSTRATIVE AIRFLOW ARITHMETIC

Internal duct	18 in. x 12 in.
Actual area	1.5 ft2
Average velocity	800 FPM
Estimated airflow	800 x 1.5 = 1,200 CFM

Illustrative only. The values are not a recommendation, acceptance criterion, code requirement, or design target.

CONCEPTUAL MULTI-POINT TRAVERSE



CONCEPT ONLY - NOT A STANDARDS POINT LAYOUT

Use the governing traverse method and point locations. Record each point. For pitot work, convert each velocity-pressure reading to velocity before averaging velocities. Then multiply average velocity by actual area.

CAPTURE HOOD CHECKLIST

1. Confirm hood size, range, and flow direction.
 2. Cover the complete outlet and establish a consistent seal.
 3. Allow readings to stabilize.
 4. Repeat as required by the procedure.
 5. Use only validated corrections or K-factors.
 6. Document terminal and test conditions.
- A hood adds resistance and may alter flow. One terminal reading does not establish total system airflow.

COMMON MEASUREMENT ERRORS

- Wrong TESP boundary
- TESP called airflow
- External duct dimensions
- One center reading
- Misaligned velocity probe
- Generic grille free area
- Instrument not zeroed
- Damaged or leaking tubing
- Internal liner ignored
- Poor traverse plane
- Pressure averaged before velocity
- Generic hood correction

INSTRUMENT CONTEXT

Hot wire	Local/low velocity; sensitive to direction, contamination, and turbulence.
Vane	Face or larger-opening velocity; watch angle, coverage, and grille effects.
Capture hood	Terminal CFM; watch seal, resistance, range, geometry, and corrections.
Psychrometer	Temperature/humidity for separate moist-air analysis, not airflow or pressure.

FROM MEASUREMENT TO ANYHVAC

Airflow & Velocity
Enter representative FPM and actual ft2 to estimate CFM. It does not validate the measurement method.

System Pressure Loss
Straight loss uses entered friction rate and length. CFM and geometry do not derive friction rate.

Friction Rate
Enter a valid design ESP allowance, component losses, and TEL. Measured TESP is not automatically the ESP input.

HVAC Duct Calculator
Enter design CFM and friction rate for sizing relationships. Results do not prove field compliance.